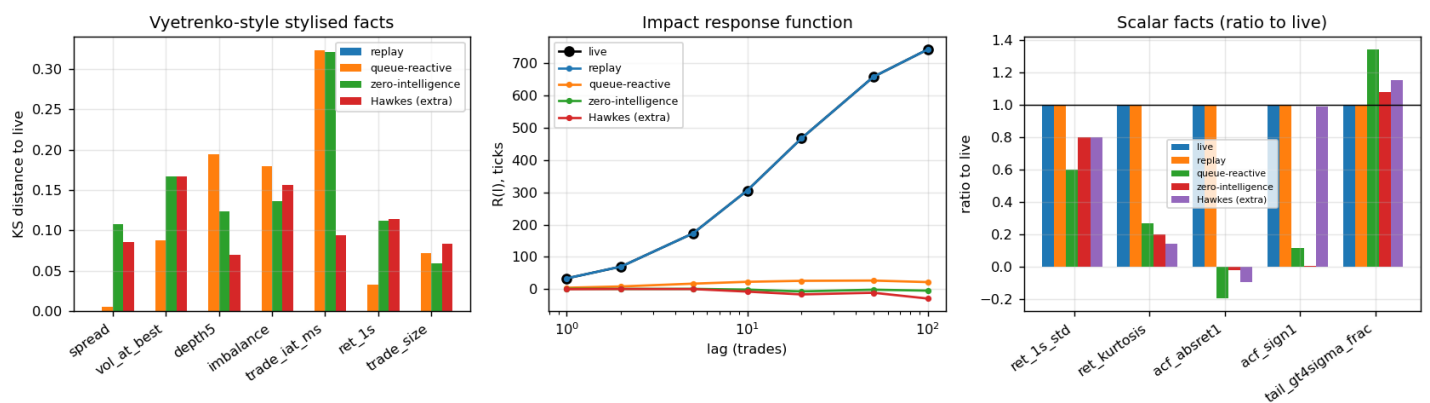


Order-Book Execution Simulator and TCA

Question. How much does the choice of order-book simulator change the ranking of execution algorithms, and how much implementation shortfall can an order-flow-aware scheduler save once queue position and fills are modelled honestly?

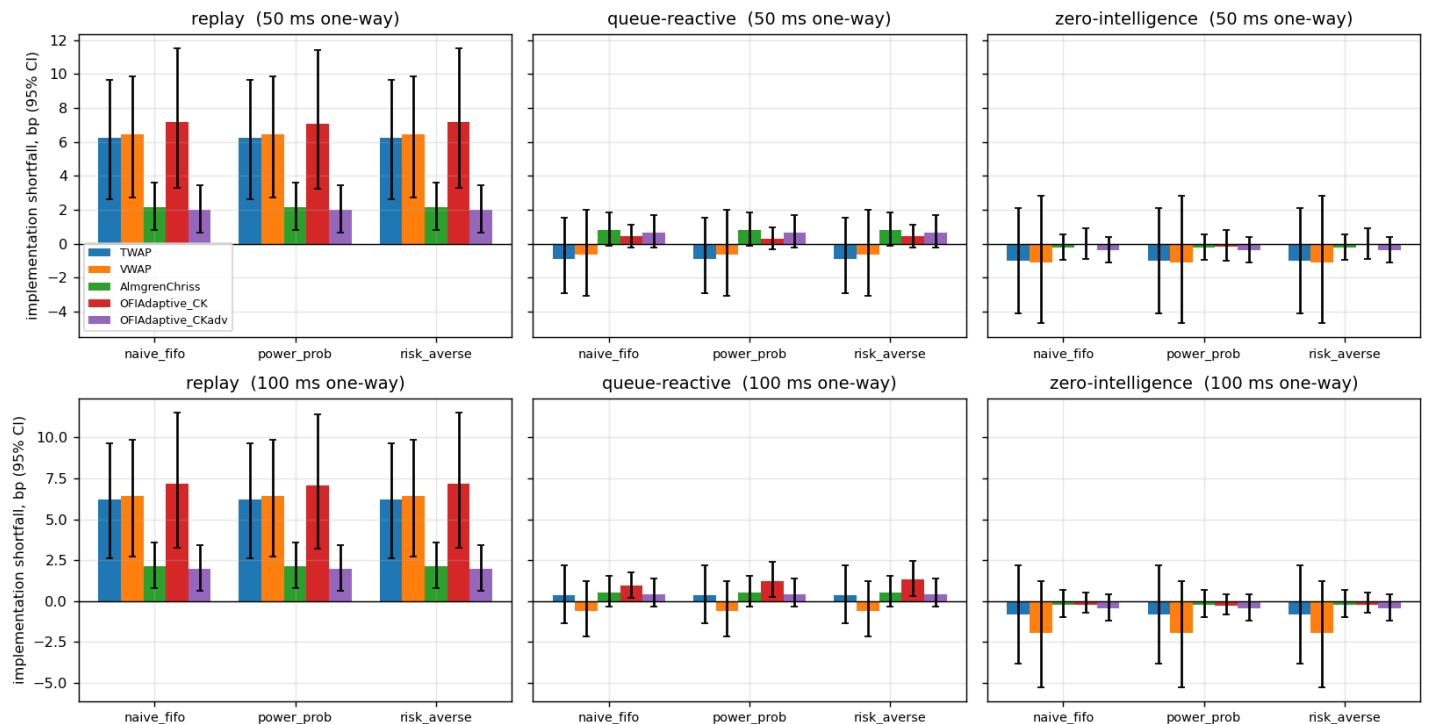
Method. A C++ event-driven engine replays self-recorded Binance BTCUSD L2 data with three queue-position models (naive FIFO, hftbacktest power-probability, risk-averse), and simulates the same market with a calibrated queue-reactive model (Huang-Lehalle-Rosenbaum 2015 with Bodor-Carlier order sizes and a data-driven reference-price jump layer) and a zero-intelligence Poisson null (Abergel-Jedidi 2013). TWAP, VWAP, Almgren-Chriss and an OFI-adaptive Cont-Kukanov scheduler run unchanged in every mode, in-process and through a FIX 4.4 gateway. Realism is scored with Vytenko-style stylised facts and the impact response function; fill probabilities and adverse selection are measured from replay probes and checked against order-level Bitstamp data; implementation shortfall is attributed into spread, timing, impact and opportunity cost with common-random-number counterfactuals.

Caveats. Crypto is a proxy for equities/FX (24/7, maker/taker fees, no last look, no internalisation). One recording day; confidence intervals over episodes within the day plus a four-day synthetic check. Coinbase's order-level channel now needs an API key, so Bitstamp's public order-level feed is the L3 ground truth.



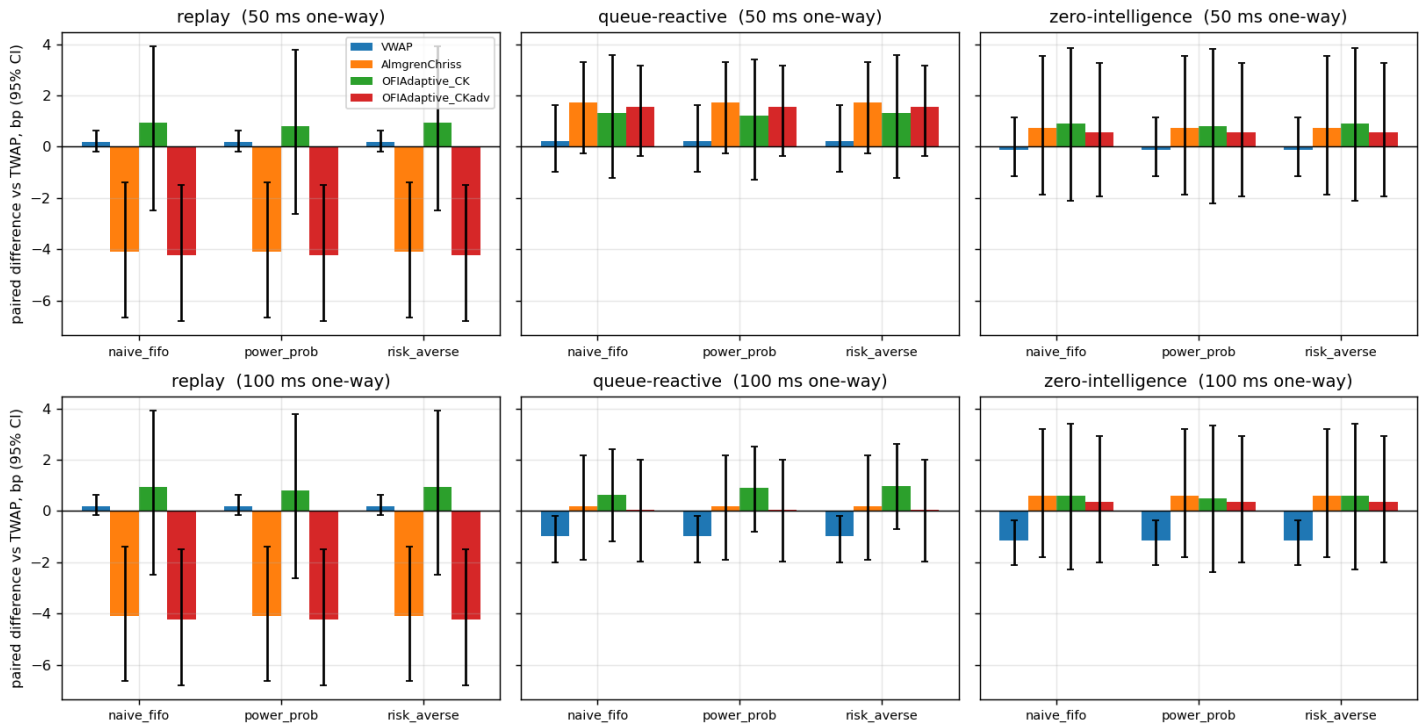
Simulator realism: KS distance per stylised fact, impact response function, scalar facts relative to live.

Cost by algorithm, simulator mode and queue assumption

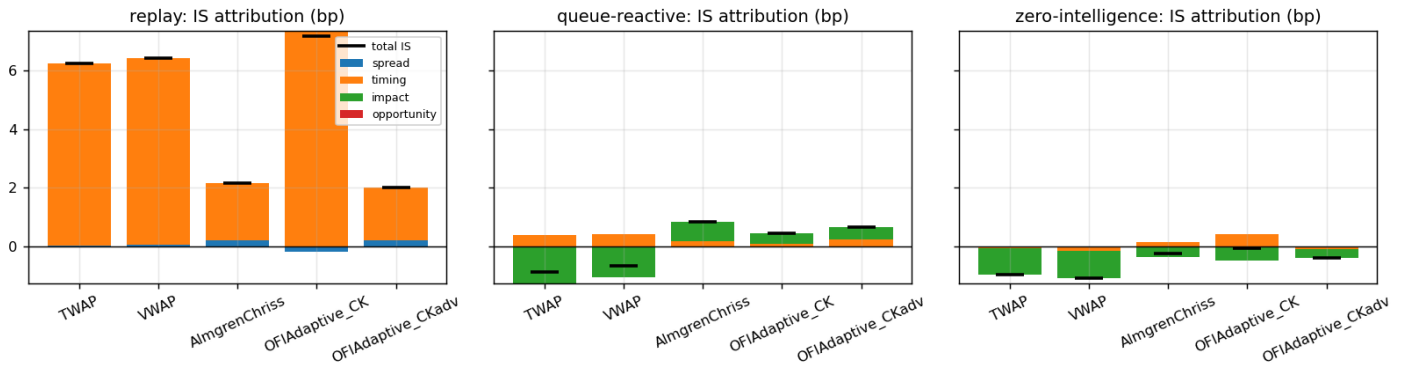


Implementation shortfall by algorithm, simulator mode and queue assumption (95% bootstrap CI).

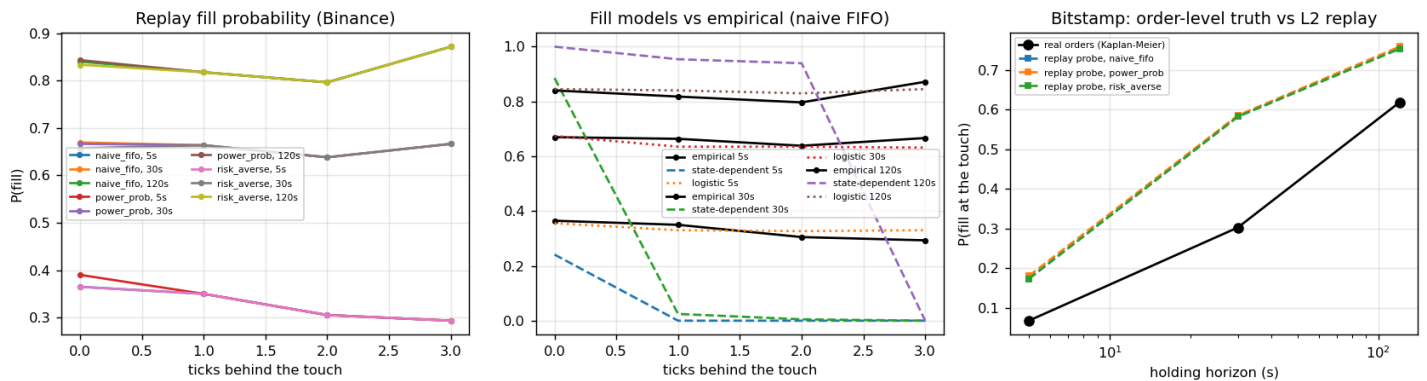
Paired per-episode cost difference vs TWAP (negative = cheaper than TWAP)



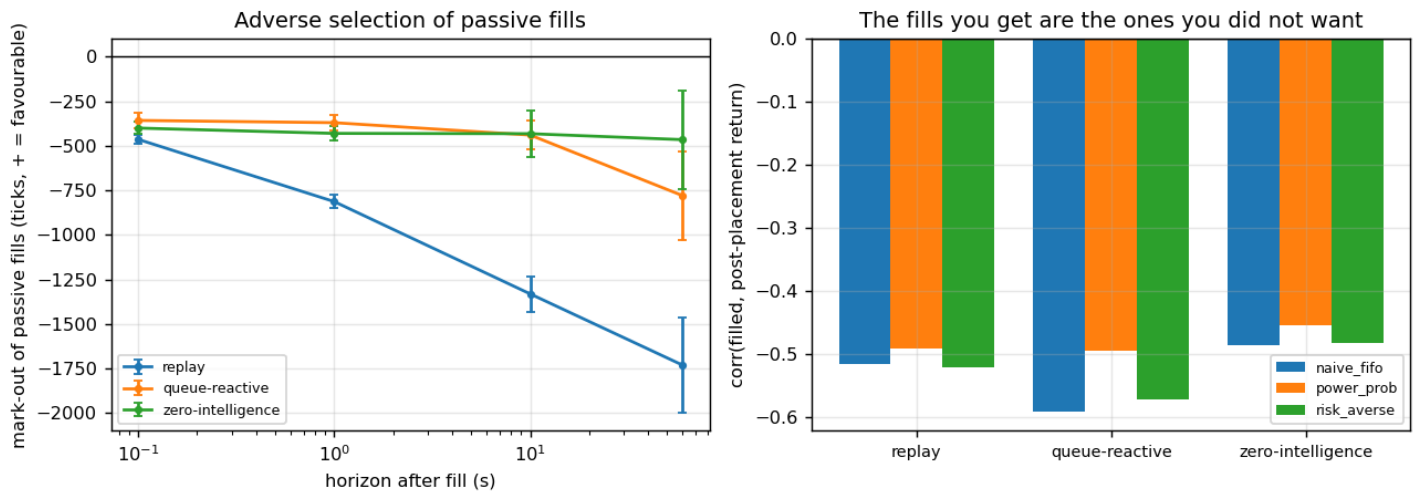
Paired difference vs TWAP; negative = cheaper than TWAP.



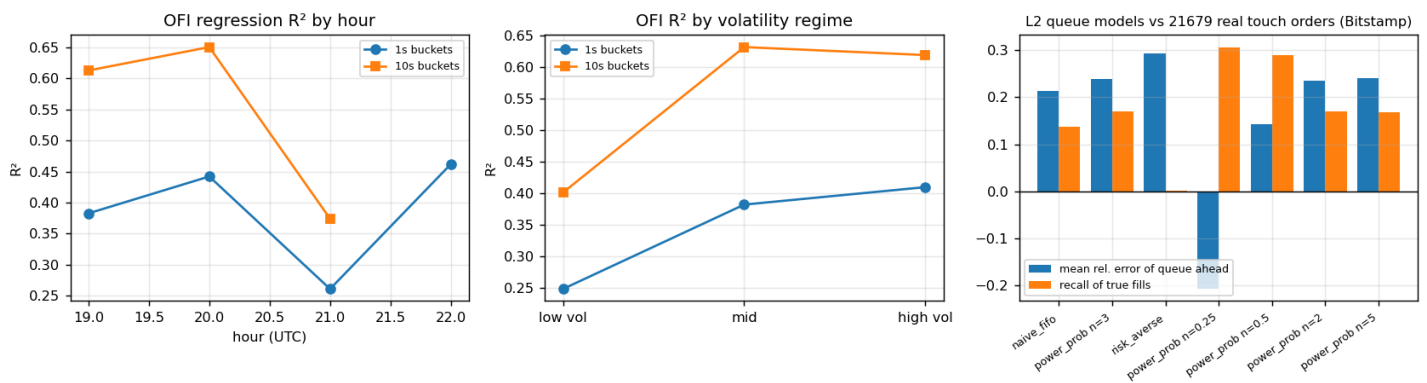
Shortfall attribution: spread, timing, impact, opportunity (sums to total by construction).



Passive fill probability: replay probes on Binance, fill models vs empirical, and Bitstamp order-level truth vs L2 replay.



Adverse selection: mark-outs of passive fills and the correlation between being filled and the post-placement return.



OFI regression R -squared by hour and volatility regime; queue-position model errors against order-level truth.

Data capture

venue	feed	duration	messages	sequence gaps	resyncs	disconnects	snapshot parity	recv - exchange time (p50 / p99)
Binance spot BTC/USD	depth@100ms + trades@100ms	2.402	448,127	2	3	0	63,248 levels, 0 mismatches	160 / 182 ms
Coinbase Exchange	live_order_book + matches@100ms	1.281 (12)	262,608	0	3	0	n/a (absolute levels)	105 / 140 ms
Bitstamp BTC/USD	live_orders + live_trades@100ms	1.133 (13, order ids)	470,571 orders, 2,691 trades	1	1	0	top-of-book vs diff_order_book: 91% agree	

Calibration (Binance)

- events decomposed: 266,372 (limit 126,289, cancel 94,854, market 45,229); trades 347,343; AES = 20252 lots; sigma(1 s) = 552 ticks; mean spread 1.15 ticks
- reference-price jumps (|dmid| >= 5 ticks between batches): 3,790; rate by touch-imbalance bucket [0.55, 0.15, 0.1, 0.11, 0.56] /s; P(up) by bucket [0.02, 0.23, 0.45, 0.77, 0.98]
- Hawkes branching ratio 0.75; OFI beta = 5.85e-04 ticks/lot, R^2 = 0.36; AC eta = 4.72e-03 ticks per lot/s

Simulator realism (live vs simulated, Binance)

statistic (KS distance to live)	replay	queue_reactive	zero_intelligence	hawkes
spread	0.00	0.01	0.11	0.09
vol_at_best	0.00	0.09	0.17	0.17
depth5	0.00	0.19	0.12	0.07
imbalance	0.00	0.18	0.14	0.16
trade_iat_ms	0.00	0.32	0.32	0.09
ret_1s	0.00	0.03	0.11	0.11
trade_size	0.00	0.07	0.06	0.08
sigma(1 s) ticks (live 555)	555	333	443	444
kurtosis of 1 s returns (live 173)	173	46.6	34.3	24.6
ACF	r	lag 1 (live 0.127)	0.127	-0.0247
trade-sign ACF lag 1 (live 0.428)	0.428	0.0488	0.00347	0.424
P(	r	> 4sigma) (live 0.00901)	0.00901	0.0121
impact response R(1)...R(100) ticks	184...743	5...22	1...-5	-0...-29

Cost ranking (Binance replay day)

parent = 300,000 lots over 300 s in 10 slices, 12 episodes, pairing unit: episode (only one day available: CI is over episodes, not days)

- latency 50 ms: 3 distinct rankings across simulator mode x queue assumption
- OFIAdaptive_CKadv < AlmgrenChriss < TWAP < VWAP < OFIAdaptive_CK in replay/naive_fifo, replay/power_prob, replay/risk_averse
- TWAP < VWAP < OFIAdaptive_CK < OFIAdaptive_CKadv < AlmgrenChriss in queue_reactive/naive_fifo, queue_reactive/power_prob, queue_reactive/risk_averse
- VWAP < TWAP < OFIAdaptive_CKadv < AlmgrenChriss < OFIAdaptive_CK in zero_intelligence/naive_fifo, zero_intelligence/power_prob, zero_intelligence/risk_averse
- latency 100 ms: 3 distinct rankings across simulator mode x queue assumption
- OFIAdaptive_CKadv < AlmgrenChriss < TWAP < VWAP < OFIAdaptive_CK in replay/naive_fifo, replay/power_prob, replay/risk_averse
- VWAP < TWAP < OFIAdaptive_CKadv < AlmgrenChriss < OFIAdaptive_CK in queue_reactive/naive_fifo, queue_reactive/power_prob, queue_reactive/risk_averse, zero_intelligence/naive_fifo, zero_intelligence/risk_averse
- VWAP < TWAP < OFIAdaptive_CKadv < OFIAdaptive_CK < AlmgrenChriss in zero_intelligence/power_prob

Latency

mode	queue mode	algo	IS bp (mean)	95% CI	vs TWAP bp	95% CI	spread	timing	impact	opp.	fees	passive %	rank
replay	naive_fifo	TWAP	6.24	[2.61, 9.65]	+0.00	[0.00, 0.00]	0.03	6.21	0.00	0.00	0.00	0%	3
replay	naive_fifo	VWAP	6.43	[2.71, 9.86]	+0.19	[-0.18, 0.63]	0.03	6.40	0.00	0.00	0.00	0%	4
replay	naive_fifo	AlmgrenChriss	8.15	[0.81, 3.61]	-4.10	[-6.64, -1.38]	0.18	1.96	0.00	0.00	0.00	0%	2
replay	naive_fifo	OFIAdaptive_CK	7.17	[3.30, 11.55]	+0.93	[-2.47, 3.93]	-0.18	7.36	0.00	0.00	0.00	94%	5
replay	naive_fifo	OFIAdaptive_CKadv	7.06	[0.67, 3.42]	-4.24	[-6.80, -1.50]	0.20	1.80	0.00	0.00	0.00	0%	1
replay	power_prob	TWAP	6.24	[2.61, 9.65]	+0.00	[0.00, 0.00]	0.03	6.21	0.00	0.00	0.00	0%	3
replay	power_prob	VWAP	6.43	[2.71, 9.86]	+0.19	[-0.18, 0.63]	0.03	6.40	0.00	0.00	0.00	0%	4
replay	power_prob	AlmgrenChriss	8.15	[0.81, 3.61]	-4.10	[-6.64, -1.38]	0.18	1.96	0.00	0.00	0.00	0%	2
replay	power_prob	OFIAdaptive_CK	7.17	[3.22, 11.41]	+0.82	[-2.62, 3.79]	-0.19	7.25	-0.01	0.00	0.00	94%	5

replay	power_prob	OFIAdaptive	0.62	[0.67, 3.42]	-4.24	[-6.80, -1.50]	0.20	1.80	0.00	0.00	0.00	0%	1
replay	risk_averse	TWAP	6.24	[2.61, 9.65]	+0.00	[0.00, 0.00]	0.03	6.21	0.00	0.00	0.00	0%	3
replay	risk_averse	VWAP	6.43	[2.71, 9.86]	+0.19	[-0.18, 0.63]	0.03	6.40	0.00	0.00	0.00	0%	4
replay	risk_averse	AlmgrenChris	8.15	[0.81, 3.61]	-4.10	[-6.64, -1.38]	0.18	1.96	0.00	0.00	0.00	0%	2
replay	risk_averse	OFIAdaptive	0.62	[3.30, 11.55]	+0.93	[-2.47, 3.93]	-0.18	7.36	0.00	0.00	0.00	94%	5
replay	risk_averse	OFIAdaptive	0.62	[0.67, 3.42]	-4.24	[-6.80, -1.50]	0.20	1.80	0.00	0.00	0.00	0%	1
queue_reactive	naive_fifo	TWAP	-0.89	[-2.91, 1.53]	+0.00	[0.00, 0.00]	0.00	0.38	-1.28	0.00	0.00	0%	1
queue_reactive	naive_fifo	VWAP	-0.67	[-3.08, 2.01]	+0.23	[-0.96, 1.62]	0.00	0.41	-1.08	0.00	0.00	0%	2
queue_reactive	naive_fifo	AlmgrenChris	8.82	[-0.13, 1.85]	+1.72	[-0.26, 3.31]	0.00	0.18	0.65	0.00	0.00	0%	5
queue_reactive	naive_fifo	OFIAdaptive	0.62	[-0.23, 1.11]	+1.32	[-1.23, 3.57]	-0.00	0.08	0.35	0.00	0.00	95%	3
queue_reactive	naive_fifo	OFIAdaptive	0.62	[0.67, 3.42]	+1.56	[-0.35, 3.17]	0.00	0.24	0.42	0.00	0.00	0%	4
queue_reactive	power_prob	TWAP	-0.89	[-2.91, 1.53]	+0.00	[0.00, 0.00]	0.00	0.38	-1.28	0.00	0.00	0%	1
queue_reactive	power_prob	VWAP	-0.67	[-3.08, 2.01]	+0.23	[-0.96, 1.62]	0.00	0.41	-1.08	0.00	0.00	0%	2
queue_reactive	power_prob	AlmgrenChris	8.82	[-0.13, 1.85]	+1.72	[-0.26, 3.31]	0.00	0.18	0.65	0.00	0.00	0%	5
queue_reactive	power_prob	OFIAdaptive	0.62	[-0.33, 0.98]	+1.20	[-1.30, 3.42]	-0.00	0.16	0.15	0.00	0.00	95%	3
queue_reactive	power_prob	OFIAdaptive	0.62	[0.67, 3.42]	+1.56	[-0.35, 3.17]	0.00	0.24	0.42	0.00	0.00	0%	4
queue_reactive	risk_averse	TWAP	-0.89	[-2.91, 1.53]	+0.00	[0.00, 0.00]	0.00	0.38	-1.28	0.00	0.00	0%	1
queue_reactive	risk_averse	VWAP	-0.67	[-3.08, 2.01]	+0.23	[-0.96, 1.62]	0.00	0.41	-1.08	0.00	0.00	0%	2
queue_reactive	risk_averse	AlmgrenChris	8.82	[-0.13, 1.85]	+1.72	[-0.26, 3.31]	0.00	0.18	0.65	0.00	0.00	0%	5
queue_reactive	risk_averse	OFIAdaptive	0.62	[-0.23, 1.10]	+1.32	[-1.22, 3.56]	-0.00	0.09	0.34	0.00	0.00	95%	3
queue_reactive	risk_averse	OFIAdaptive	0.62	[0.67, 3.42]	+1.56	[-0.35, 3.17]	0.00	0.24	0.42	0.00	0.00	0%	4
zero_intelligence	naive_fifo	TWAP	-0.98	[-4.13, 2.08]	+0.00	[0.00, 0.00]	0.00	-0.08	-0.91	0.00	0.00	0%	2
zero_intelligence	naive_fifo	VWAP	-1.09	[-4.69, 2.82]	-0.11	[-1.16, 1.14]	0.00	-0.15	-0.94	0.00	0.00	0%	1
zero_intelligence	naive_fifo	AlmgrenChris	8.24	[-0.96, 0.56]	+0.74	[-1.88, 3.54]	0.00	0.13	-0.37	0.00	0.00	0%	4
zero_intelligence	naive_fifo	OFIAdaptive	0.62	[-0.90, 0.91]	+0.92	[-2.10, 3.85]	-0.00	0.42	-0.49	0.00	0.00	100%	5
zero_intelligence	naive_fifo	OFIAdaptive	0.62	[-1.12, 0.41]	+0.58	[-1.95, 3.27]	0.00	-0.09	-0.31	0.00	0.00	0%	3
zero_intelligence	power_prob	TWAP	-0.98	[-4.13, 2.08]	+0.00	[0.00, 0.00]	0.00	-0.08	-0.91	0.00	0.00	0%	2
zero_intelligence	power_prob	VWAP	-1.09	[-4.69, 2.82]	-0.11	[-1.16, 1.14]	0.00	-0.15	-0.94	0.00	0.00	0%	1
zero_intelligence	power_prob	AlmgrenChris	8.24	[-0.96, 0.56]	+0.74	[-1.88, 3.54]	0.00	0.13	-0.37	0.00	0.00	0%	4
zero_intelligence	power_prob	OFIAdaptive	0.62	[-1.00, 0.82]	+0.82	[-2.22, 3.81]	-0.00	0.37	-0.54	0.00	0.00	100%	5
zero_intelligence	power_prob	OFIAdaptive	0.62	[-1.12, 0.41]	+0.58	[-1.95, 3.27]	0.00	-0.09	-0.31	0.00	0.00	0%	3
zero_intelligence	risk_averse	TWAP	-0.98	[-4.13, 2.08]	+0.00	[0.00, 0.00]	0.00	-0.08	-0.91	0.00	0.00	0%	2
zero_intelligence	risk_averse	VWAP	-1.09	[-4.69, 2.82]	-0.11	[-1.16, 1.14]	0.00	-0.15	-0.94	0.00	0.00	0%	1
zero_intelligence	risk_averse	AlmgrenChris	8.24	[-0.96, 0.56]	+0.74	[-1.88, 3.54]	0.00	0.13	-0.37	0.00	0.00	0%	4
zero_intelligence	risk_averse	OFIAdaptive	0.62	[-0.90, 0.91]	+0.92	[-2.10, 3.85]	-0.00	0.42	-0.49	0.00	0.00	100%	5
zero_intelligence	risk_averse	OFIAdaptive	0.62	[-1.12, 0.41]	+0.58	[-1.95, 3.27]	0.00	-0.09	-0.31	0.00	0.00	0%	3

Latency 100 ms one-way

mode	queue model	algo	IS bp (mean)	95% CI	vs TWAP bp	95% CI	spread	timing	impact	opp.	fees	passive %	rank
replay	naive_fifo	TWAP	6.24	[2.61, 9.64]	+0.00	[0.00, 0.00]	0.03	6.22	-0.00	0.00	0.00	0%	3
replay	naive_fifo	VWAP	6.43	[2.70, 9.86]	+0.19	[-0.17, 0.63]	0.03	6.40	-0.00	0.00	0.00	0%	4
replay	naive_fifo	AlmgrenChris	8.14	[0.78, 3.60]	-4.10	[-6.64, -1.40]	0.18	1.97	-0.01	0.00	0.00	0%	2
replay	naive_fifo	OFIAdaptive	0.62	[3.28, 11.55]	+0.93	[-2.48, 3.93]	-0.19	7.36	0.00	0.00	0.00	93%	5
replay	naive_fifo	OFIAdaptive	0.62	[0.66, 3.42]	-4.24	[-6.80, -1.50]	0.20	1.81	-0.01	0.00	0.00	0%	1
replay	power_prob	TWAP	6.24	[2.61, 9.64]	+0.00	[0.00, 0.00]	0.03	6.22	-0.00	0.00	0.00	0%	3
replay	power_prob	VWAP	6.43	[2.70, 9.86]	+0.19	[-0.17, 0.63]	0.03	6.40	-0.00	0.00	0.00	0%	4
replay	power_prob	AlmgrenChris	8.14	[0.78, 3.60]	-4.10	[-6.64, -1.40]	0.18	1.97	-0.01	0.00	0.00	0%	2
replay	power_prob	OFIAdaptive	0.62	[3.22, 11.40]	+0.82	[-2.62, 3.79]	-0.19	7.25	-0.01	0.00	0.00	93%	5
replay	power_prob	OFIAdaptive	0.62	[0.66, 3.42]	-4.24	[-6.80, -1.50]	0.20	1.81	-0.01	0.00	0.00	0%	1
replay	risk_averse	TWAP	6.24	[2.61, 9.64]	+0.00	[0.00, 0.00]	0.03	6.22	-0.00	0.00	0.00	0%	3
replay	risk_averse	VWAP	6.43	[2.70, 9.86]	+0.19	[-0.17, 0.63]	0.03	6.40	-0.00	0.00	0.00	0%	4
replay	risk_averse	AlmgrenChris	8.14	[0.78, 3.60]	-4.10	[-6.64, -1.40]	0.18	1.97	-0.01	0.00	0.00	0%	2
replay	risk_averse	OFIAdaptive	0.62	[3.28, 11.55]	+0.93	[-2.47, 3.93]	-0.19	7.36	0.00	0.00	0.00	93%	5
replay	risk_averse	OFIAdaptive	0.62	[0.66, 3.42]	-4.24	[-6.80, -1.50]	0.20	1.81	-0.01	0.00	0.00	0%	1
queue_reactive	naive_fifo	TWAP	0.36	[-1.36, 2.21]	+0.00	[0.00, 0.00]	0.00	0.40	-0.04	0.00	0.00	0%	2
queue_reactive	naive_fifo	VWAP	-0.62	[-2.18, 1.21]	-0.97	[-2.00, -0.19]	0.00	0.43	-1.04	0.00	0.00	0%	1
queue_reactive	naive_fifo	AlmgrenChris	8.54	[-0.31, 1.53]	+0.18	[-1.90, 2.18]	0.00	0.17	0.37	0.00	0.00	0%	4
queue_reactive	naive_fifo	OFIAdaptive	0.62	[0.22, 1.78]	+0.61	[-1.17, 2.41]	-0.00	0.12	0.85	0.00	0.00	92%	5

queue_reactive	naive_fifo	OFIAdaptive	0Kadv	[-0.36, 1.40]	+0.06	[-1.96, 1.99]	0.00	0.16	0.25	0.00	0.00	0%	3
queue_reactive	power_prob	TWAP	0.36	[-1.36, 2.21]	+0.00	[0.00, 0.00]	0.00	0.40	-0.04	0.00	0.00	0%	2
queue_reactive	power_prob	VWAP	-0.62	[-2.18, 1.21]	-0.97	[-2.00, -0.19]	0.00	0.43	-1.04	0.00	0.00	0%	1
queue_reactive	power_prob	AlmgrenChrs	\$0.54	[-0.31, 1.53]	+0.18	[-1.90, 2.18]	0.00	0.17	0.37	0.00	0.00	0%	4
queue_reactive	power_prob	OFIAdaptive	1K25	[0.25, 2.39]	+0.89	[-0.80, 2.52]	-0.00	0.05	1.19	0.00	0.00	94%	5
queue_reactive	power_prob	OFIAdaptive	0Kadv	[-0.36, 1.40]	+0.06	[-1.96, 1.99]	0.00	0.16	0.25	0.00	0.00	0%	3
queue_reactive	risk_averse	TWAP	0.36	[-1.36, 2.21]	+0.00	[0.00, 0.00]	0.00	0.40	-0.04	0.00	0.00	0%	2
queue_reactive	risk_averse	VWAP	-0.62	[-2.18, 1.21]	-0.97	[-2.00, -0.19]	0.00	0.43	-1.04	0.00	0.00	0%	1
queue_reactive	risk_averse	AlmgrenChrs	\$0.54	[-0.31, 1.53]	+0.18	[-1.90, 2.18]	0.00	0.17	0.37	0.00	0.00	0%	4
queue_reactive	risk_averse	OFIAdaptive	1K1	[0.30, 2.44]	+0.96	[-0.72, 2.63]	0.00	0.06	1.25	0.00	0.00	92%	5
queue_reactive	risk_averse	OFIAdaptive	0Kadv	[-0.36, 1.40]	+0.06	[-1.96, 1.99]	0.00	0.16	0.25	0.00	0.00	0%	3
zero_intelligence	naive_fifo	TWAP	-0.80	[-3.83, 2.19]	+0.00	[0.00, 0.00]	0.00	-0.03	-0.77	0.00	0.00	0%	2
zero_intelligence	naive_fifo	VWAP	-1.96	[-5.27, 1.22]	-1.16	[-2.10, -0.36]	0.00	-0.19	-1.78	0.00	0.00	0%	1
zero_intelligence	naive_fifo	AlmgrenChrs	\$0.21	[-0.97, 0.68]	+0.60	[-1.82, 3.21]	0.00	0.30	-0.50	0.00	0.00	0%	4
zero_intelligence	naive_fifo	OFIAdaptive	0K20	[-0.73, 0.50]	+0.60	[-2.28, 3.39]	-0.00	0.23	-0.43	0.00	0.00	100%	5
zero_intelligence	naive_fifo	OFIAdaptive	0Kadv	[-1.19, 0.40]	+0.36	[-1.99, 2.92]	0.00	0.05	-0.49	0.00	0.00	0%	3
zero_intelligence	power_prob	TWAP	-0.80	[-3.83, 2.19]	+0.00	[0.00, 0.00]	0.00	-0.03	-0.77	0.00	0.00	0%	2
zero_intelligence	power_prob	VWAP	-1.96	[-5.27, 1.22]	-1.16	[-2.10, -0.36]	0.00	-0.19	-1.78	0.00	0.00	0%	1
zero_intelligence	power_prob	AlmgrenChrs	\$0.21	[-0.97, 0.68]	+0.60	[-1.82, 3.21]	0.00	0.30	-0.50	0.00	0.00	0%	5
zero_intelligence	power_prob	OFIAdaptive	0K80	[-0.83, 0.42]	+0.50	[-2.38, 3.35]	-0.00	0.18	-0.47	0.00	0.00	100%	4
zero_intelligence	power_prob	OFIAdaptive	0Kadv	[-1.19, 0.40]	+0.36	[-1.99, 2.92]	0.00	0.05	-0.49	0.00	0.00	0%	3
zero_intelligence	risk_averse	TWAP	-0.80	[-3.83, 2.19]	+0.00	[0.00, 0.00]	0.00	-0.03	-0.77	0.00	0.00	0%	2
zero_intelligence	risk_averse	VWAP	-1.96	[-5.27, 1.22]	-1.16	[-2.10, -0.36]	0.00	-0.19	-1.78	0.00	0.00	0%	1
zero_intelligence	risk_averse	AlmgrenChrs	\$0.21	[-0.97, 0.68]	+0.60	[-1.82, 3.21]	0.00	0.30	-0.50	0.00	0.00	0%	4
zero_intelligence	risk_averse	OFIAdaptive	0K20	[-0.73, 0.50]	+0.60	[-2.28, 3.39]	-0.00	0.23	-0.43	0.00	0.00	100%	5
zero_intelligence	risk_averse	OFIAdaptive	0Kadv	[-1.19, 0.40]	+0.36	[-1.99, 2.92]	0.00	0.05	-0.49	0.00	0.00	0%	3

Cost ranking on 4 synthetic queue-reactive days (day-paired CIs)

- latency 50 ms: 2 distinct rankings
- latency 100 ms: 3 distinct rankings

Latency

mode	queue mode	algo	IS bp (mean)	95% CI	vs TWAP bp	95% CI	spread	timing	impact	opp.	fees	passive %	rank
replay	naive_fifo	TWAP	0.65	[-0.73, 2.15]	+0.00	[0.00, 0.00]	0.00	0.65	0.00	0.00	0.00	0%	4
replay	naive_fifo	VWAP	0.65	[-0.73, 2.15]	+0.00	[0.00, 0.00]	0.00	0.65	0.00	0.00	0.00	0%	5
replay	naive_fifo	AlmgrenChrs	\$0.00	[-0.44, 0.48]	-0.65	[-1.67, 0.32]	0.00	0.00	0.00	0.00	0.00	0%	2
replay	naive_fifo	OFIAdaptive	0K19	[0.29, 0.81]	-0.06	[-1.37, 1.04]	0.48	0.10	0.00	0.00	0.00	100%	3
replay	naive_fifo	OFIAdaptive	0Kadv	[-0.48, 0.41]	-0.71	[-1.74, 0.25]	0.00	-0.06	0.00	0.00	0.00	0%	1
replay	power_prob	TWAP	0.65	[-0.73, 2.15]	+0.00	[0.00, 0.00]	0.00	0.65	0.00	0.00	0.00	0%	4
replay	power_prob	VWAP	0.65	[-0.73, 2.15]	+0.00	[0.00, 0.00]	0.00	0.65	0.00	0.00	0.00	0%	5
replay	power_prob	AlmgrenChrs	\$0.00	[-0.44, 0.48]	-0.65	[-1.67, 0.32]	0.00	0.00	0.00	0.00	0.00	0%	2
replay	power_prob	OFIAdaptive	0K6	[0.13, 0.78]	-0.20	[-1.45, 0.87]	0.43	0.05	-0.02	0.00	0.00	100%	3
replay	power_prob	OFIAdaptive	0Kadv	[-0.48, 0.41]	-0.71	[-1.74, 0.25]	0.00	-0.06	0.00	0.00	0.00	0%	1
replay	risk_averse	TWAP	0.65	[-0.73, 2.15]	+0.00	[0.00, 0.00]	0.00	0.65	0.00	0.00	0.00	0%	4
replay	risk_averse	VWAP	0.65	[-0.73, 2.15]	+0.00	[0.00, 0.00]	0.00	0.65	0.00	0.00	0.00	0%	5
replay	risk_averse	AlmgrenChrs	\$0.00	[-0.44, 0.48]	-0.65	[-1.67, 0.32]	0.00	0.00	0.00	0.00	0.00	0%	2
replay	risk_averse	OFIAdaptive	0K60	[0.30, 0.83]	-0.05	[-1.34, 1.04]	0.49	0.11	0.00	0.00	0.00	100%	3
replay	risk_averse	OFIAdaptive	0Kadv	[-0.48, 0.41]	-0.71	[-1.74, 0.25]	0.00	-0.06	0.00	0.00	0.00	0%	1
queue_reactive	naive_fifo	TWAP	-0.20	[-1.53, 1.13]	+0.00	[0.00, 0.00]	0.00	0.34	-0.54	0.00	0.00	0%	1
queue_reactive	naive_fifo	VWAP	-0.20	[-1.53, 1.13]	+0.00	[0.00, 0.00]	0.00	0.34	-0.54	0.00	0.00	0%	2
queue_reactive	naive_fifo	AlmgrenChrs	\$0.14	[-0.26, 0.45]	+0.35	[-0.96, 1.50]	0.00	-0.09	0.23	0.00	0.00	0%	4
queue_reactive	naive_fifo	OFIAdaptive	0K7	[0.54, 1.44]	+1.17	[0.02, 2.33]	-0.00	-0.30	1.27	0.00	0.00	100%	5
queue_reactive	naive_fifo	OFIAdaptive	0Kadv	[-0.40, 0.46]	+0.26	[-1.13, 1.48]	0.00	-0.10	0.16	0.00	0.00	0%	3
queue_reactive	power_prob	TWAP	-0.20	[-1.53, 1.13]	+0.00	[0.00, 0.00]	0.00	0.34	-0.54	0.00	0.00	0%	1
queue_reactive	power_prob	VWAP	-0.20	[-1.53, 1.13]	+0.00	[0.00, 0.00]	0.00	0.34	-0.54	0.00	0.00	0%	2
queue_reactive	power_prob	AlmgrenChrs	\$0.14	[-0.26, 0.45]	+0.35	[-0.96, 1.50]	0.00	-0.09	0.23	0.00	0.00	0%	4
queue_reactive	power_prob	OFIAdaptive	1K2	[0.62, 1.89]	+1.33	[0.29, 2.36]	-0.00	-0.43	1.55	0.00	0.00	100%	5
queue_reactive	power_prob	OFIAdaptive	0Kadv	[-0.40, 0.46]	+0.26	[-1.13, 1.48]	0.00	-0.10	0.16	0.00	0.00	0%	3

queue_reactive_risk_averse	naive_fifo	TWAP	-0.20	[-1.53, 1.13]	+0.00	[0.00, 0.00]	0.00	0.34	-0.54	0.00	0.00	0%	1
queue_reactive_risk_averse	naive_fifo	VWAP	-0.20	[-1.53, 1.13]	+0.00	[0.00, 0.00]	0.00	0.34	-0.54	0.00	0.00	0%	2
queue_reactive_risk_averse	naive_fifo	AlmgrenChriss	-0.14	[-0.26, 0.45]	+0.35	[-0.96, 1.50]	0.00	-0.09	0.23	0.00	0.00	0%	4
queue_reactive_risk_averse	naive_fifo	OFIAdaptive_C1	0.87	[0.55, 1.44]	+1.17	[0.03, 2.31]	-0.00	-0.30	1.27	0.00	0.00	100%	5
queue_reactive_risk_averse	naive_fifo	OFIAdaptive_C1Adv	0.66	[-0.40, 0.46]	+0.26	[-1.13, 1.48]	0.00	-0.10	0.16	0.00	0.00	0%	3
zero_intelligence_naive_fifo	naive_fifo	TWAP	-1.68	[-3.65, 0.26]	+0.00	[0.00, 0.00]	0.00	-2.19	0.50	0.00	0.00	0%	1
zero_intelligence_naive_fifo	naive_fifo	VWAP	-1.68	[-3.65, 0.26]	+0.00	[0.00, 0.00]	0.00	-2.19	0.50	0.00	0.00	0%	2
zero_intelligence_naive_fifo	naive_fifo	AlmgrenChriss	-0.24	[-0.59, 0.32]	+1.45	[0.05, 3.06]	0.00	-0.16	-0.08	0.00	0.00	0%	4
zero_intelligence_naive_fifo	naive_fifo	OFIAdaptive_C1	0.89	[-0.23, 1.01]	+2.07	[0.26, 3.70]	-0.00	0.11	0.28	0.00	0.00	99%	5
zero_intelligence_naive_fifo	naive_fifo	OFIAdaptive_C1Adv	0.22	[-0.78, 0.28]	+1.40	[0.02, 2.86]	0.00	-0.26	-0.02	0.00	0.00	0%	3
zero_intelligence_power_prob	naive_fifo	TWAP	-1.68	[-3.65, 0.26]	+0.00	[0.00, 0.00]	0.00	-2.19	0.50	0.00	0.00	0%	1
zero_intelligence_power_prob	naive_fifo	VWAP	-1.68	[-3.65, 0.26]	+0.00	[0.00, 0.00]	0.00	-2.19	0.50	0.00	0.00	0%	2
zero_intelligence_power_prob	naive_fifo	AlmgrenChriss	-0.24	[-0.59, 0.32]	+1.45	[0.05, 3.06]	0.00	-0.16	-0.08	0.00	0.00	0%	4
zero_intelligence_power_prob	naive_fifo	OFIAdaptive_C1	0.82	[-0.28, 0.96]	+2.01	[0.15, 3.66]	-0.00	0.04	0.28	0.00	0.00	99%	5
zero_intelligence_power_prob	naive_fifo	OFIAdaptive_C1Adv	0.22	[-0.78, 0.28]	+1.40	[0.02, 2.86]	0.00	-0.26	-0.02	0.00	0.00	0%	3
zero_intelligence_risk_averse	naive_fifo	TWAP	-1.68	[-3.65, 0.26]	+0.00	[0.00, 0.00]	0.00	-2.19	0.50	0.00	0.00	0%	1
zero_intelligence_risk_averse	naive_fifo	VWAP	-1.68	[-3.65, 0.26]	+0.00	[0.00, 0.00]	0.00	-2.19	0.50	0.00	0.00	0%	2
zero_intelligence_risk_averse	naive_fifo	AlmgrenChriss	-0.24	[-0.59, 0.32]	+1.45	[0.05, 3.06]	0.00	-0.16	-0.08	0.00	0.00	0%	4
zero_intelligence_risk_averse	naive_fifo	OFIAdaptive_C1	0.87	[-0.23, 0.97]	+2.05	[0.24, 3.70]	-0.00	0.13	0.24	0.00	0.00	99%	5
zero_intelligence_risk_averse	naive_fifo	OFIAdaptive_C1Adv	0.22	[-0.78, 0.28]	+1.40	[0.02, 2.86]	0.00	-0.26	-0.02	0.00	0.00	0%	3

Latency 100 ms one-way

mode	queue model	algo	IS bp (mean)	95% CI	vs TWAP bp	95% CI	spread	timing	impact	opp.	fees	passive %	rank
replay	naive_fifo	TWAP	0.66	[-0.73, 2.16]	+0.00	[0.00, 0.00]	0.00	0.66	0.00	0.00	0.00	0%	4
replay	naive_fifo	VWAP	0.66	[-0.73, 2.16]	+0.00	[0.00, 0.00]	0.00	0.66	0.00	0.00	0.00	0%	5
replay	naive_fifo	AlmgrenChriss	-0.08	[-0.43, 0.60]	-0.59	[-1.56, 0.34]	0.00	0.07	0.00	0.00	0.00	0%	2
replay	naive_fifo	OFIAdaptive_C1	0.59	[0.29, 0.81]	-0.07	[-1.38, 1.03]	0.48	0.10	0.00	0.00	0.00	100%	3
replay	naive_fifo	OFIAdaptive_C1Adv	0.66	[-0.47, 0.53]	-0.65	[-1.63, 0.27]	0.00	0.01	0.00	0.00	0.00	0%	1
replay	power_prob	TWAP	0.66	[-0.73, 2.16]	+0.00	[0.00, 0.00]	0.00	0.66	0.00	0.00	0.00	0%	4
replay	power_prob	VWAP	0.66	[-0.73, 2.16]	+0.00	[0.00, 0.00]	0.00	0.66	0.00	0.00	0.00	0%	5
replay	power_prob	AlmgrenChriss	-0.08	[-0.43, 0.60]	-0.59	[-1.56, 0.34]	0.00	0.07	0.00	0.00	0.00	0%	2
replay	power_prob	OFIAdaptive_C1	0.46	[0.13, 0.78]	-0.20	[-1.45, 0.86]	0.43	0.05	-0.02	0.00	0.00	100%	3
replay	power_prob	OFIAdaptive_C1Adv	0.66	[-0.47, 0.53]	-0.65	[-1.63, 0.27]	0.00	0.01	0.00	0.00	0.00	0%	1
replay	risk_averse	TWAP	0.66	[-0.73, 2.16]	+0.00	[0.00, 0.00]	0.00	0.66	0.00	0.00	0.00	0%	4
replay	risk_averse	VWAP	0.66	[-0.73, 2.16]	+0.00	[0.00, 0.00]	0.00	0.66	0.00	0.00	0.00	0%	5
replay	risk_averse	AlmgrenChriss	-0.08	[-0.43, 0.60]	-0.59	[-1.56, 0.34]	0.00	0.07	0.00	0.00	0.00	0%	2
replay	risk_averse	OFIAdaptive_C1	0.60	[0.30, 0.83]	-0.06	[-1.35, 1.03]	0.49	0.11	0.00	0.00	0.00	100%	3
replay	risk_averse	OFIAdaptive_C1Adv	0.66	[-0.47, 0.53]	-0.65	[-1.63, 0.27]	0.00	0.01	0.00	0.00	0.00	0%	1
queue_reactive_naive_fifo	naive_fifo	TWAP	0.97	[0.65, 1.28]	+0.00	[0.00, 0.00]	0.00	0.35	0.61	0.00	0.00	0%	3
queue_reactive_naive_fifo	naive_fifo	VWAP	0.97	[0.65, 1.28]	+0.00	[0.00, 0.00]	0.00	0.35	0.61	0.00	0.00	0%	4
queue_reactive_naive_fifo	naive_fifo	AlmgrenChriss	-0.08	[-0.46, 0.30]	-1.05	[-1.39, -0.49]	0.00	-0.04	-0.04	0.00	0.00	0%	2
queue_reactive_naive_fifo	naive_fifo	OFIAdaptive_C1	0.98	[0.21, 1.56]	+0.01	[-0.59, 0.60]	-0.00	-0.02	1.00	0.00	0.00	98%	5
queue_reactive_naive_fifo	naive_fifo	OFIAdaptive_C1Adv	0.45	[-0.52, 0.25]	-1.12	[-1.56, -0.50]	0.00	-0.02	-0.14	0.00	0.00	0%	1
queue_reactive_power_prob	power_prob	TWAP	0.97	[0.65, 1.28]	+0.00	[0.00, 0.00]	0.00	0.35	0.61	0.00	0.00	0%	3
queue_reactive_power_prob	power_prob	VWAP	0.97	[0.65, 1.28]	+0.00	[0.00, 0.00]	0.00	0.35	0.61	0.00	0.00	0%	4
queue_reactive_power_prob	power_prob	AlmgrenChriss	-0.08	[-0.46, 0.30]	-1.05	[-1.39, -0.49]	0.00	-0.04	-0.04	0.00	0.00	0%	2
queue_reactive_power_prob	power_prob	OFIAdaptive_C1	1.00	[0.58, 1.59]	+0.14	[-0.36, 0.63]	-0.00	0.07	1.04	0.00	0.00	99%	5
queue_reactive_power_prob	power_prob	OFIAdaptive_C1Adv	0.45	[-0.52, 0.25]	-1.12	[-1.56, -0.50]	0.00	-0.02	-0.14	0.00	0.00	0%	1
queue_reactive_risk_averse	risk_averse	TWAP	0.97	[0.65, 1.28]	+0.00	[0.00, 0.00]	0.00	0.35	0.61	0.00	0.00	0%	4
queue_reactive_risk_averse	risk_averse	VWAP	0.97	[0.65, 1.28]	+0.00	[0.00, 0.00]	0.00	0.35	0.61	0.00	0.00	0%	5
queue_reactive_risk_averse	risk_averse	AlmgrenChriss	-0.08	[-0.46, 0.30]	-1.05	[-1.39, -0.49]	0.00	-0.04	-0.04	0.00	0.00	0%	2
queue_reactive_risk_averse	risk_averse	OFIAdaptive_C1	0.87	[0.18, 1.55]	-0.00	[-0.61, 0.59]	-0.00	-0.00	0.97	0.00	0.00	98%	3
queue_reactive_risk_averse	risk_averse	OFIAdaptive_C1Adv	0.45	[-0.52, 0.25]	-1.12	[-1.56, -0.50]	0.00	-0.02	-0.14	0.00	0.00	0%	1
zero_intelligence_naive_fifo	naive_fifo	TWAP	-1.95	[-3.68, -0.22]	+0.00	[0.00, 0.00]	0.00	-2.21	0.26	0.00	0.00	0%	1
zero_intelligence_naive_fifo	naive_fifo	VWAP	-1.95	[-3.68, -0.22]	+0.00	[0.00, 0.00]	0.00	-2.21	0.26	0.00	0.00	0%	2
zero_intelligence_naive_fifo	naive_fifo	AlmgrenChriss	-0.16	[-0.63, 0.33]	+1.78	[0.52, 3.05]	0.00	-0.09	-0.07	0.00	0.00	0%	4
zero_intelligence_naive_fifo	naive_fifo	OFIAdaptive_C1	0.87	[-0.41, 1.15]	+2.32	[1.31, 3.29]	-0.00	0.09	0.28	0.00	0.00	100%	5
zero_intelligence_naive_fifo	naive_fifo	OFIAdaptive_C1Adv	0.22	[-0.81, 0.32]	+1.70	[0.54, 2.87]	0.00	-0.18	-0.07	0.00	0.00	0%	3

zero_intelligence	power_prob	TWAP	-1.95	[-3.68, -0.22]	+0.00	[0.00, 0.00]	0.00	-2.21	0.26	0.00	0.00	0%	1
zero_intelligence	power_prob	VWAP	-1.95	[-3.68, -0.22]	+0.00	[0.00, 0.00]	0.00	-2.21	0.26	0.00	0.00	0%	2
zero_intelligence	power_prob	AlmgrenChriss	-0.16	[-0.63, 0.33]	+1.78	[0.52, 3.05]	0.00	-0.09	-0.07	0.00	0.00	0%	4
zero_intelligence	power_prob	OFIAdaptive	0.20	[-0.47, 0.93]	+2.14	[0.86, 3.26]	-0.00	-0.06	0.26	0.00	0.00	99%	5
zero_intelligence	power_prob	OFIAdaptive	-0.24	[-0.81, 0.32]	+1.70	[0.54, 2.87]	0.00	-0.18	-0.07	0.00	0.00	0%	3
zero_intelligence	risk_averse	TWAP	-1.95	[-3.68, -0.22]	+0.00	[0.00, 0.00]	0.00	-2.21	0.26	0.00	0.00	0%	1
zero_intelligence	risk_averse	VWAP	-1.95	[-3.68, -0.22]	+0.00	[0.00, 0.00]	0.00	-2.21	0.26	0.00	0.00	0%	2
zero_intelligence	risk_averse	AlmgrenChriss	-0.16	[-0.63, 0.33]	+1.78	[0.52, 3.05]	0.00	-0.09	-0.07	0.00	0.00	0%	4
zero_intelligence	risk_averse	OFIAdaptive	0.19	[-0.40, 1.19]	+2.34	[1.31, 3.31]	-0.00	0.07	0.33	0.00	0.00	100%	5
zero_intelligence	risk_averse	OFIAdaptive	-0.24	[-0.81, 0.32]	+1.70	[0.54, 2.87]	0.00	-0.18	-0.07	0.00	0.00	0%	3

Passive fill probability (Binance replay probes)

queue model	horizon	P(fill) at touch	+1 tick	+2	+3	touch by queue tercile (low/mid/high)
naive_fifo	5 s	0.36	0.35	0.31	0.29	0.73/0.62/0.53
naive_fifo	30 s	0.67	0.66	0.64	0.67	0.73/0.62/0.53
naive_fifo	120 s	0.84	0.82	0.80	0.87	0.73/0.62/0.53
power_prob	5 s	0.39	0.35	0.31	0.29	0.71/0.62/0.56
power_prob	30 s	0.67	0.66	0.64	0.67	0.71/0.62/0.56
power_prob	120 s	0.84	0.82	0.80	0.87	0.71/0.62/0.56
risk_averse	5 s	0.36	0.35	0.31	0.29	0.71/0.61/0.53
risk_averse	30 s	0.66	0.66	0.64	0.67	0.71/0.61/0.53
risk_averse	120 s	0.83	0.82	0.80	0.87	0.71/0.61/0.53

Fill models vs empirical at the touch (naive FIFO): 5s: emp 0.36, state-dependent 0.24, logistic 0.36, Poisson 0.60; 30s: emp 0.67, state-dependent 0.89, logistic 0.67, Poisson 0.90; 120s: emp 0.84, state-dependent 1.00, logistic 0.85, Poisson 0.97

Order-level check of the L2 queue models (Bitstamp)

21,687 real orders created at the touch, tracked to fill/cancel; median size 0.1020 BTC.

naive_fifo	+0.21	0.32	0.14	0.000
power_prob (n=3)	+0.24	0.32	0.17	0.011
risk_averse	+0.29	0.31	0.00	0.000
power_prob (n=0.25)	-0.21	0.71	0.31	0.023
power_prob (n=0.5)	+0.14	0.37	0.29	0.018
power_prob (n=2)	+0.24	0.31	0.17	0.008
power_prob (n=5)	+0.24	0.32	0.17	0.011

Kaplan-Meier fill probability of real touch orders (cancellation = censoring): 5s 0.07, 30s 0.30, 120s 0.62

Replay probes at the touch on the same tape: naive_fifo: 5s 0.17, 30s 0.58, 120s 0.75; power_prob: 5s 0.18, 30s 0.59, 120s 0.76; risk_averse: 5s 0.17, 30s 0.58, 120s 0.75

Adverse selection of passive fills (mark-outs, ticks, + = favourable)

mode	queue model	P(fill)	100 ms	1 s	10 s	60 s	corr(filled, post-placement return)
replay	naive_fifo	0.67	-464 +/- 24	-813 +/- 38	-1332 +/- 100	-1732 +/- 266	-0.51
replay	power_prob	0.68	-414 +/- 28	-740 +/- 42	-1216 +/- 103	-1544 +/- 267	-0.49
replay	risk_averse	0.66	-477 +/- 24	-831 +/- 38	-1376 +/- 100	-1788 +/- 267	-0.52
queue_reactive	naive_fifo	0.79	-357 +/- 42	-370 +/- 45	-439 +/- 82	-780 +/- 248	-0.59
queue_reactive	power_prob	0.85	-263 +/- 46	-287 +/- 49	-318 +/- 86	-592 +/- 236	-0.49
queue_reactive	risk_averse	0.79	-372 +/- 42	-389 +/- 45	-426 +/- 82	-711 +/- 247	-0.57
zero_intelligence	naive_fifo	0.88	-400 +/- 35	-430 +/- 40	-432 +/- 132	-465 +/- 278	-0.49
zero_intelligence	power_prob	0.89	-281 +/- 39	-292 +/- 46	-300 +/- 126	-404 +/- 272	-0.45
zero_intelligence	risk_averse	0.89	-403 +/- 35	-432 +/- 39	-428 +/- 131	-490 +/- 275	-0.48

Queue-position value (Moallemi-Yuan, measured): P(fill) x (half-spread + 60 s mark-out), by tercile of queue ahead at placement, naive FIFO:

replay	0	34,317	0.83	-1698	-1417.5
replay	1	155,180	0.62	-1053	-655.7
replay	2	463,995	0.54	-2564	-1384.4
queue_reactive	0	75,083	0.82	-996	-816.3
queue_reactive	1	194,052	0.76	-304	-230.4

queue_reactive	2	420,329	0.78	-1017	-792.8
zero_intelligence	0	38,551	0.82	-443	-362.7
zero_intelligence	1	188,808	0.96	-519	-497.4
zero_intelligence	2	381,611	0.86	-427	-366.9

OFI regression (Cont-Kukanov-Stoikov)

- 1s buckets: $\beta = 5.85e-04$ ticks/lot, $R^2 = 0.36$, $n = 10,079$; by hour UTC: 19h 0.38, 20h 0.44, 21h 0.26, 22h 0.46; by volatility tercile: 0.25, 0.38, 0.41

- 10s buckets: $\beta = 6.93e-04$ ticks/lot, $R^2 = 0.57$, $n = 1,008$; by hour UTC: 19h 0.61, 20h 0.65, 21h 0.37; by volatility tercile: 0.40, 0.63, 0.62

Known-answer tests and FIX self-test

KNOWN-ANSWER TESTS PASSED: 52/52 checks

TWAP in-process: 49 fills, IS +18.815 bp | FIX: 49 fills, IS +18.815 bp | IDENTICAL
VWAP in-process: 49 fills, IS +18.815 bp | FIX: 49 fills, IS +18.815 bp | IDENTICAL
AlmgrenChriss in-process: 111 fills, IS +4.895 bp | FIX: 111 fills, IS +4.895 bp | IDENTICAL
OFIAdaptive_CK in-process: 120 fills, IS +6.432 bp | FIX: 120 fills, IS +6.432 bp | IDENTICAL
OFIAdaptive_CKadv in-process: 111 fills, IS +4.925 bp | FIX: 111 fills, IS +4.925 bp | IDENTICAL
FIX SELF-TEST PASSED: identical fills in-process and through FIX 4.4